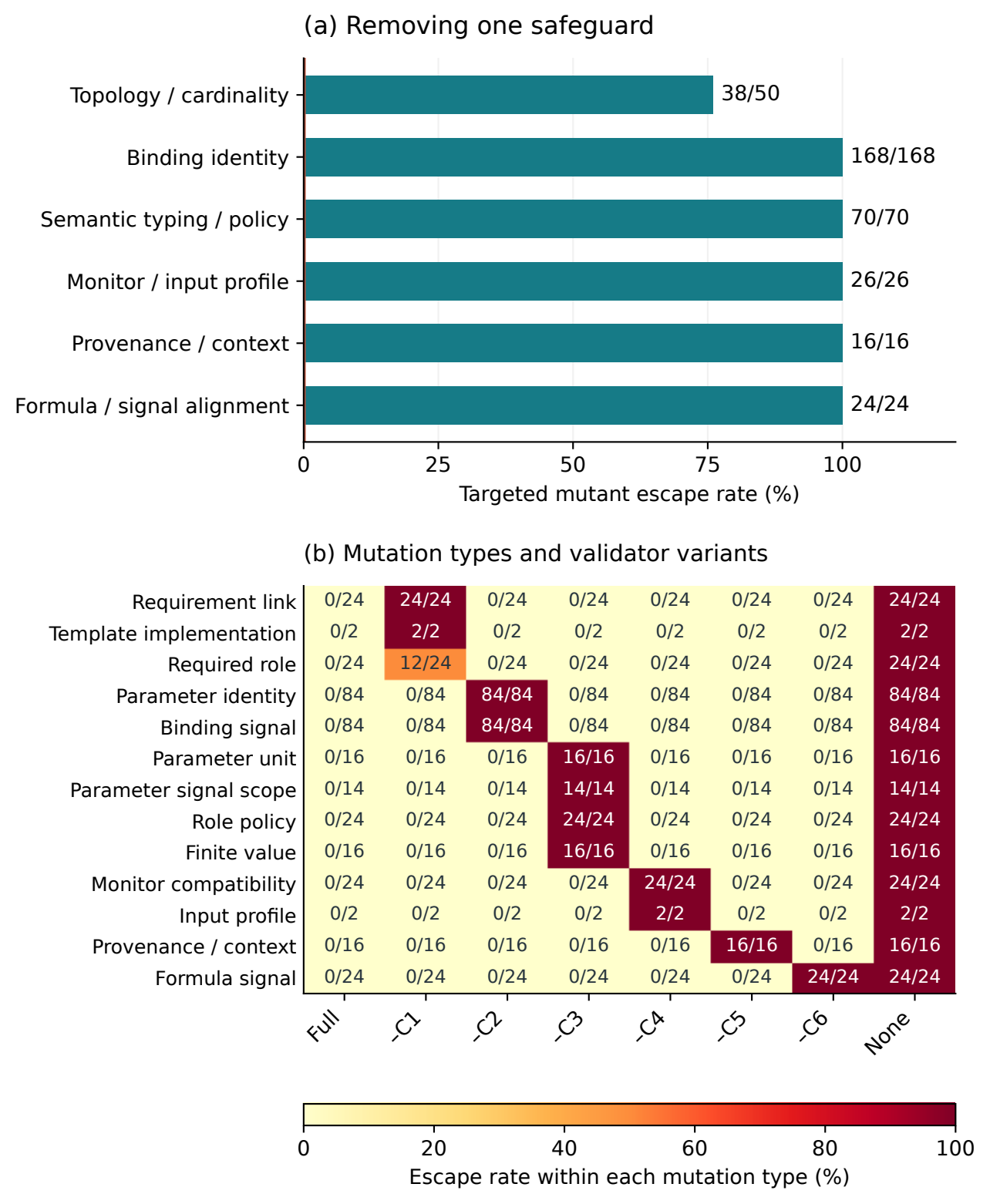

(a) Removing one safeguard
Topology / cardinality
Binding identity
Semantic typing / policy
Monitor / input profile
Provenance / context
Formula / signal alignment
38/50
168/168
70/70
26/26
16/16
24/24
0
25
50
75
100
Targeted mutant escape rate (%)
(b) Mutation types and validator variants
Requirement link 0/24 24/24 0/24 0/24 0/24 0/24 0/24 24/24
Template implementation 0/2 2/2 0/2 0/2 0/2 0/2 0/2 2/2
Required role 0/24 12/24 0/24 0/24 0/24 0/24 0/24 24/24
Parameter identity 0/84 0/84 84/84 0/84 0/84 0/84 0/84 84/84
Binding signal 0/84 0/84 84/84 0/84 0/84 0/84 0/84 84/84
Parameter unit 0/16 0/16 0/16 16/16 0/16 0/16 0/16 16/16
Parameter signal scope 0/14 0/14 0/14 14/14 0/14 0/14 0/14 14/14
Role policy 0/24 0/24 0/24 24/24 0/24 0/24 0/24 24/24
Finite value 0/16 0/16 0/16 16/16 0/16 0/16 0/16 16/16
Monitor compatibility 0/24 0/24 0/24 0/24 24/24 0/24 0/24 24/24
Input profile 0/2 0/2 0/2 0/2 2/2 0/2 0/2 2/2
Provenance / context 0/16 0/16 0/16 0/16 0/16 16/16 0/16 16/16
Formula signal 0/24 0/24 0/24 0/24 0/24 0/24 24/24 24/24
Full
−C1
−C2
−C3
−C4
−C5
−C6
None
0
20
40
60
80
100
Escape rate within each mutation type (%)